

QR Adapter

1. Product Overview



- IoT gateway
- Specific design to suitable with M18 Milwaukee Power tool
- Main function:
 - o Read QR code from Tool/Product.
 - o Convert the power for Tool/Product.
 - o Transfer Data to other device via Wi-Fi.
 - o Receive the control data from out side through Socket TCP.
 - o Control machine by IO port.

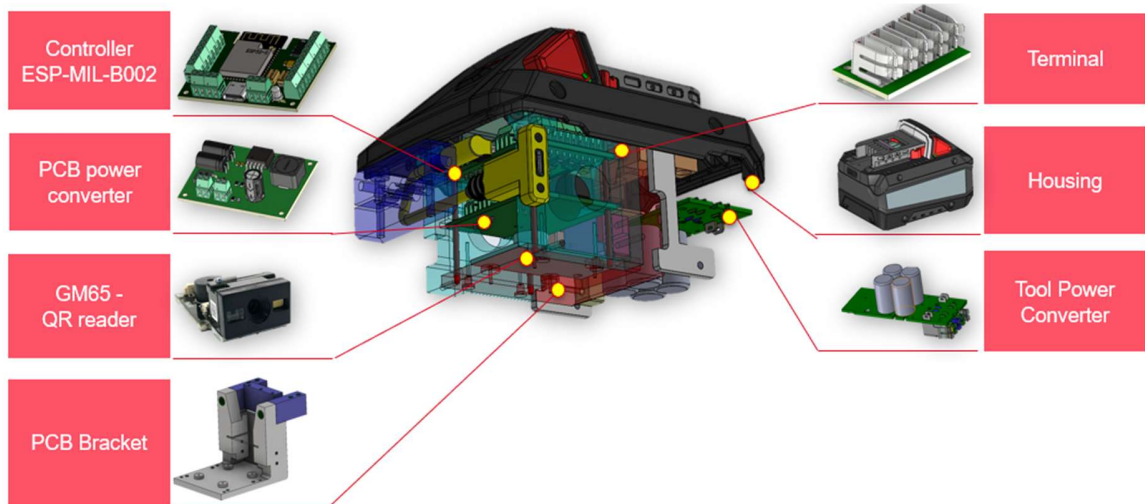
2. General Specifications:

Category	Specification
Device Type	IoT Gateway – Customized QR Adapter for M18 Tool series
Power Input	18VDC (compatible with M18 battery or equivalent external power source)
Power Output	18VDC passthrough or regulated output for Tool powering (custom design)
Scanner	GM65 2D QR Code Scanner – TTL/USB/Serial interface
Connectivity	Wi-Fi 802.11 b/g/n (2.4GHz)
Communication	TCP Socket (client/server configurable), optional UART
Control I/O	2 x Digital Output (Relay or Open Collector), 1 x Digital Input
LED Indicators	Power, Wi-Fi Status, Scan Status, Communication
Enclosure	Industrial-grade ABS/PC housing, designed to fit into M18 Tool Dock
Operating Temp	-10°C to +60°C

Dimensions	L143 x W90 x H85 (mm)
Mounting	Snap-fit, Screw Mount or Slide-in (depending on application)
Firmware	Custom embedded firmware with OTA (over-the-air) update support
Security	WPA2 Wi-Fi encryption, optional IP whitelist for socket connection
Certification	Designed for CE/EMC compliance (final validation TBD)

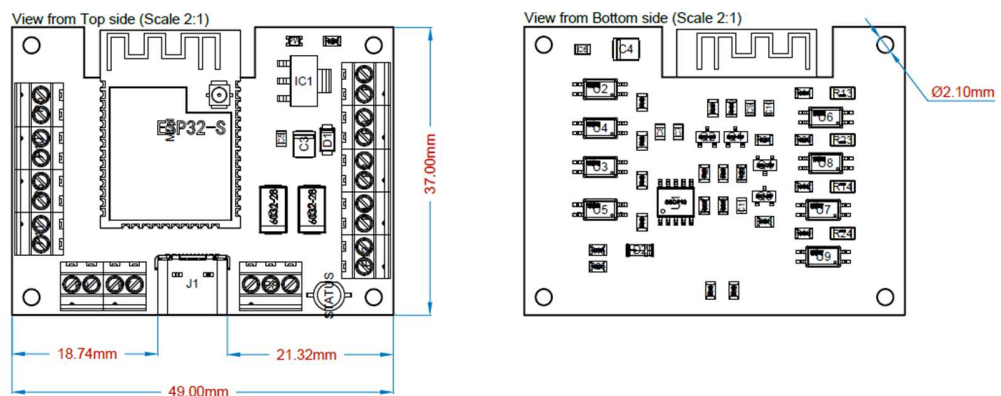
3. Hardware Architecture.

3.1 Bill of Material (BOM).

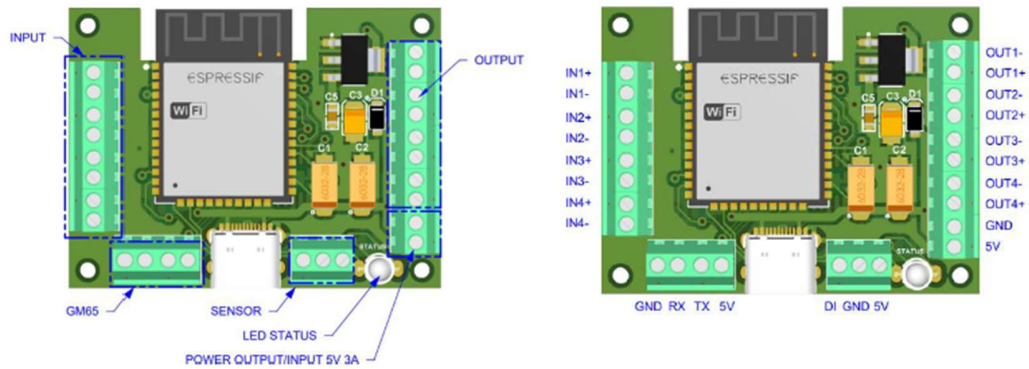


a. Controller Board

- Board Name: **ESP-MIL-Bxxx**
- xxx: Version of development.
- Latest Version: **002**



ESP-MIL-B002 Dimension



ESP-MIL-B002 pinout

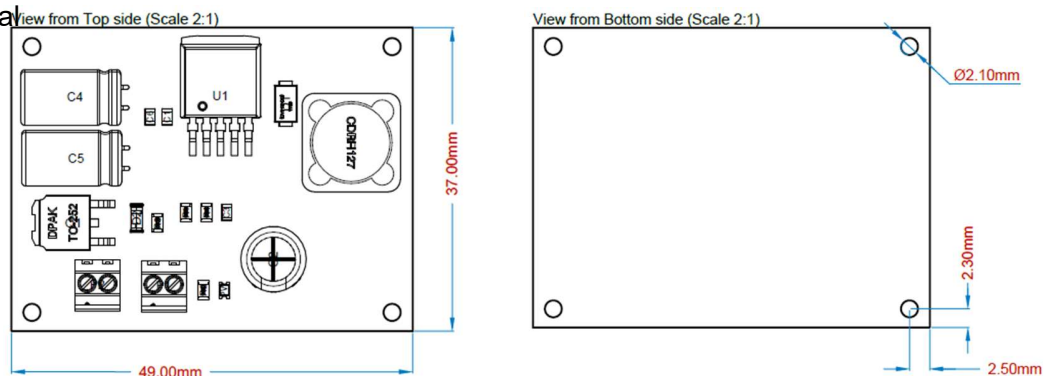
- Input/Output Layout:

No	Item	ESP-MIL-B002	ESP32	Description
1	Input	Input 1	IO36	Active in Low level
2		Input 2	IO25	
3		Input 3	IO27	
4		Input 4	IO14	
5	Output	Output 1	IO23	Active in Low level
6		Output 2	IO22	
7		Output 3	IO18	
8		Output 4	IO4	
9	GM65	TX	IO16/RX2	TX_GM65 connect to TX
10		RX	IO17/TX2	RX_GM65 connect to RX
11		Power Control	IO5	Active in High level
12	Sensor	DI	IO13	Active in Low level
13	LED Status	LED Green	IO33	LED ON in Low level
14		LED Red	IO36	LED ON in Low level

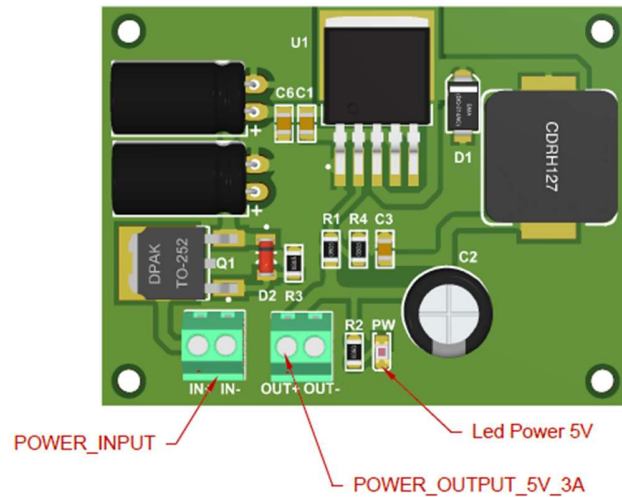
b. PCB power converter.

- Have the same dimension with Controller

- VDC converter: Convert Power of power supplier to Controller board and peripheral



PCB Dimension



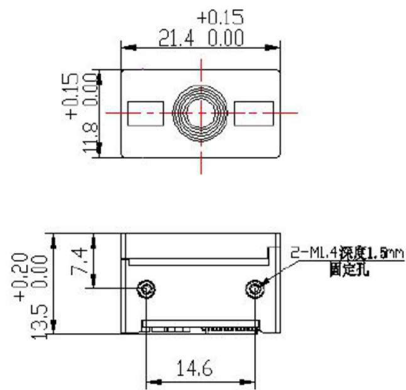
PCB power supply pinout

c. GM65 – QR reader.

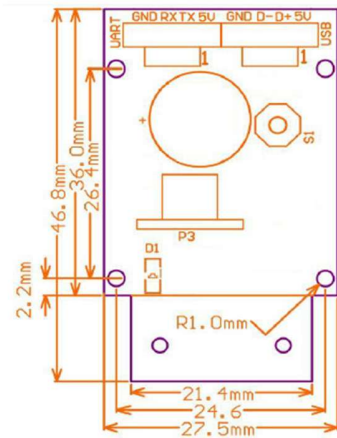
- Specification

No	Item	Spec
1	Operating Voltage	DC 4.2 - 6.0V
2	Standby Current	30mA
3	Operating Current	160mA
4	Sleep Current	3mA
5	Light	White light
6	Capture light	Red
7	Resolution	648x 488
8	Scanning angle	35° (Inclination), 28° (Elevation)
9	Interface	+ USB, UART, USB VCom + TTL - 232: - Verification: None - Data bit: 8 - Stop bit: 1 - CTSRST: No - Baud rate: adjustable, default- 9600

- Dimension (mm):

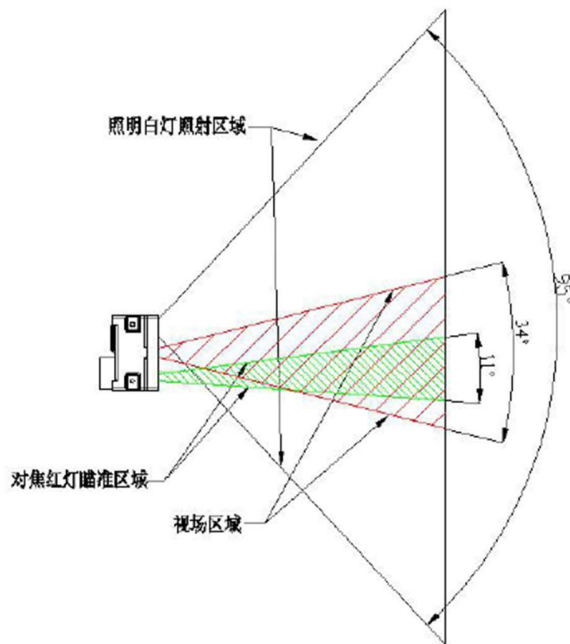


Reader



Control Panel

- Scan area (*testing in office -250 lux*).



- Configuration to adapter with QR Adapter.

Use GM65 to scan configuration QR code as follows:



Reset



Series Output



Default Continuous Mode



1500ms

Picture 1

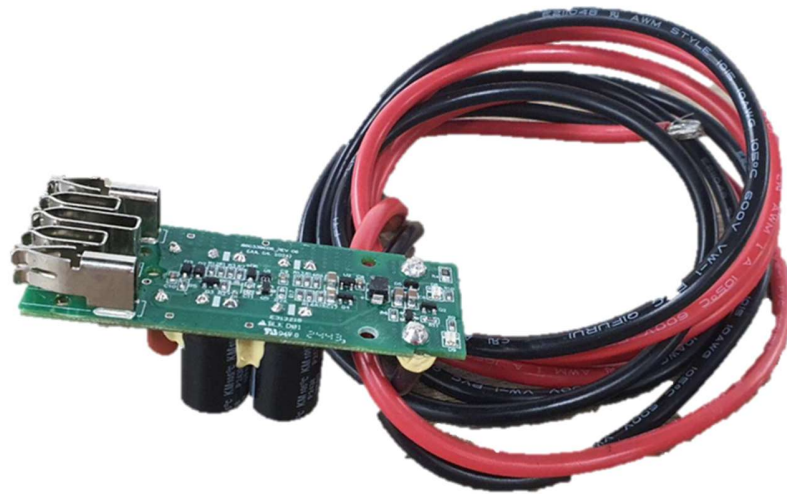
Picture 2

Picture 3

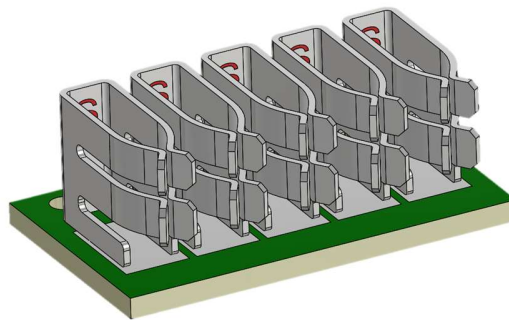
Picture 4

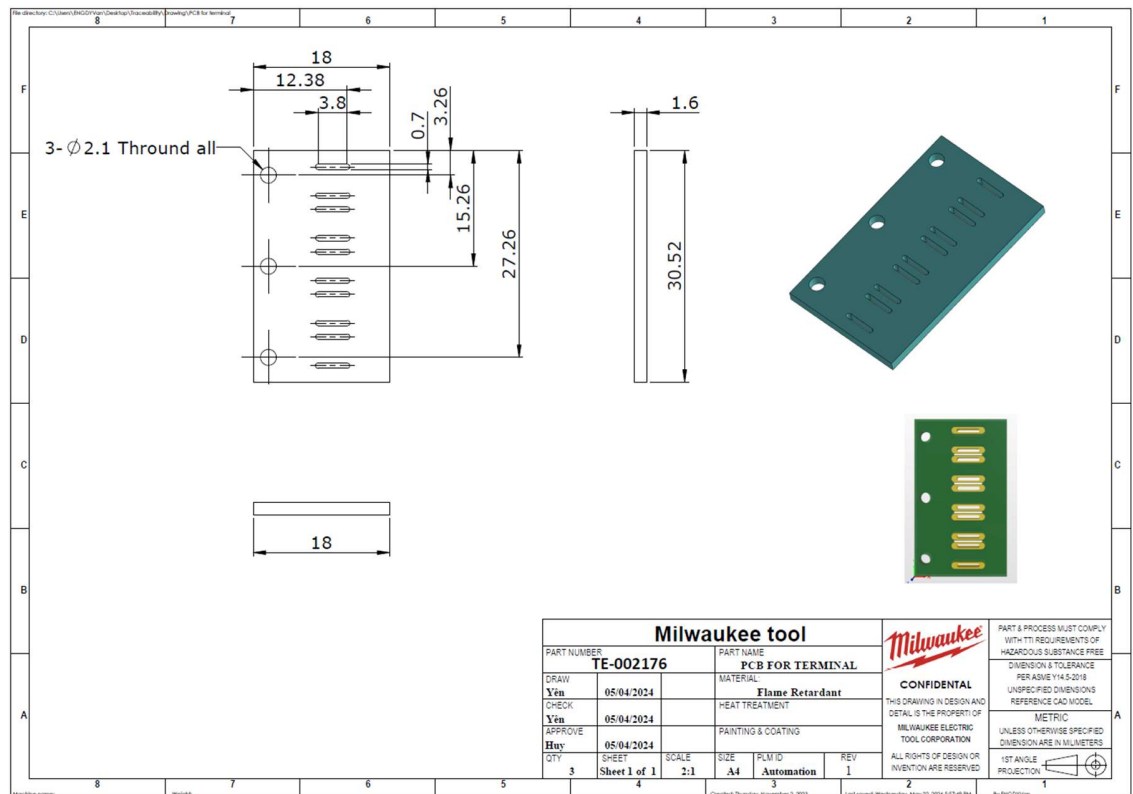
No	Item	Description
1	Picture 1	Reset factory Parameter -> waiting for the long sound response
2	Picture 2	Select module as Serial output mode
3	Picture 3	Continuously read QR mode
4	Picture 4	Sleep time between 2 times continuous

d. Tool power converter.



e. Terminal Pin.

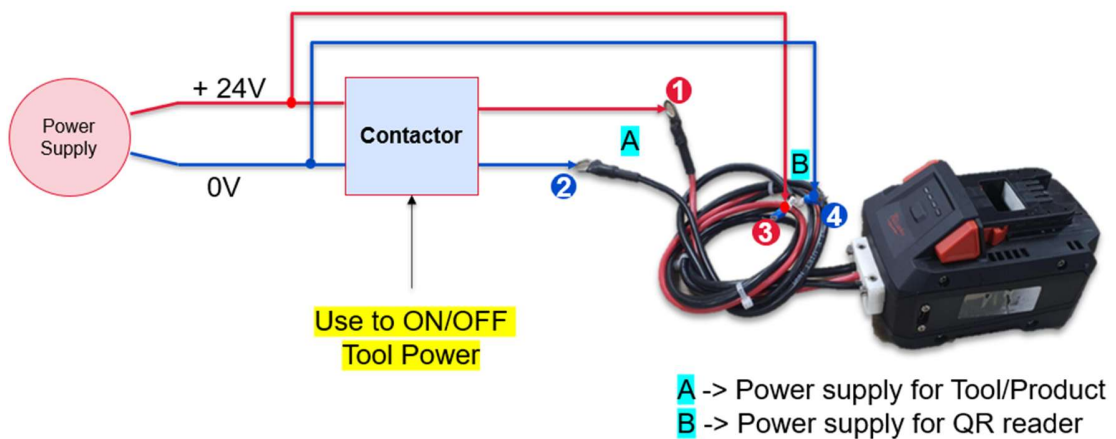




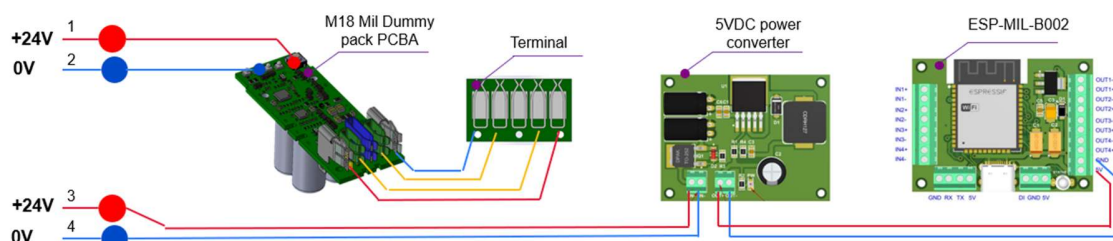
f. Housing.

g. Other mechanical part.

3.2 PCB Block Diagrams.



Power Supply for QR Adapter



Wiring diagrams

3.3 Mechanical Design.

a. Explored View

b.

3.4 Assembly Guide.

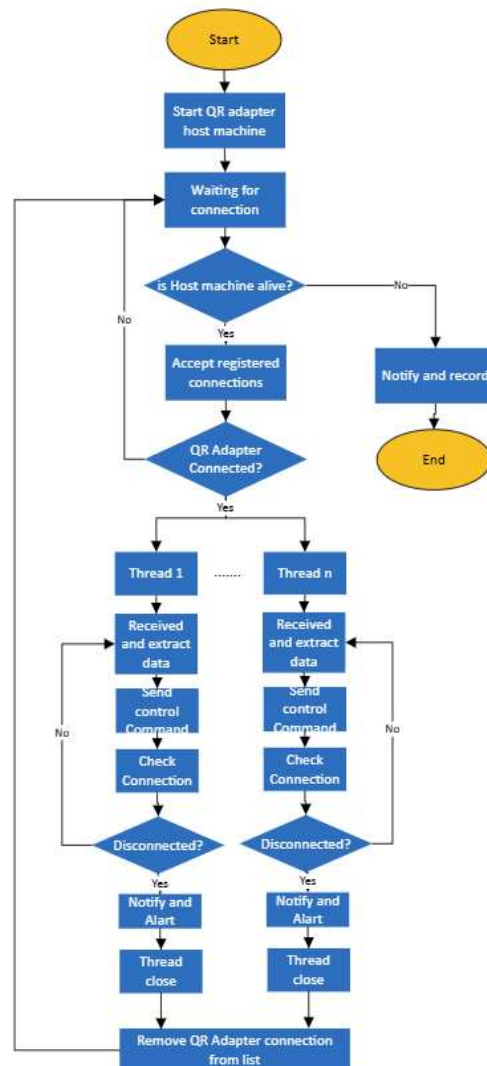
3.5 Manufacturing Notes.

4. Functionality.

4.1. Các chức năng chính.

5. Firmware

4.1. Operation diagrams.



4.2. Version control.

The version number format a.b.c.d is a four-level versioning scheme commonly used in enterprise software, embedded systems, and industrial firmware. It is an extension of Semantic Versioning (SemVer), with an additional fourth component to track build or revision information.

- **a (Major):** Introduces breaking changes that are not backward-compatible. Increments when the software undergoes significant structural changes or API redesign.
- **b (Minor):** Adds new features in a backward-compatible manner. Increments when functionality is extended without altering existing APIs.
- **c (Patch):** Fixes bugs or makes small backward-compatible improvements. Increments when minor corrections are made.
- **d (Build/Revision):** Represents internal build numbers, deployment revisions, or CI/CD build IDs. This value may increment automatically with each build to uniquely identify the output.

Example: 1.2.0.5

- **1** – Major version: initial public release
- **2** – Minor version: added new features
- **0** – Patch: no bug fixes yet
- **5** – Build: 5th internal build or deployment of this version

4.2. Download firmware guide.

6. Communication Interfaces & Protocols

5.1 Physical Interfaces.

5.2 Supported Protocols.*

7. Data Format & Configuration.

6.1 Format of configuration command.

6.2 Command to control Adapter.

6.3 Command to get data from Adapter.

8. System Integration Guide

9. Maintenance & Troubleshooting.

10. Testing & Validation

11. Appendices